

Muñoz Rodríguez Iván

Postdoc at CEA Paris-Saclay



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SCIENTIFIC PROFILE

I am an astrophysicist with expertise in the growth of supermassive black holes (SMBHs), the evolution of galaxies and Active Galactic Nuclei (AGN) across cosmic time and different environments, as well as X-ray cluster surveys. My expertise bridges observations, simulations, and statistical modelling to improve how galaxies, AGN, and galaxy clusters are identified, characterised, and used as cosmological probes. I specialise in X-ray data analysis, cluster-detection algorithms, semi-empirical modelling, and large-scale simulations.

POSTDOCTORAL POSITIONS

POSTDOCTORAL RESEARCHER | POSTDOC

January 2025 – Current | CEA Paris-Saclay

EDUCATION

PHD IN ASTRONOMY | PHD THESIS: SEMI-EMPIRICAL MODELS OF X-RAY AGN IN GALAXY CLUSTERS

October 2020 – August 2024 | National Observatory of Athens (Greece) and University of Southampton (United Kingdom)

MSC IN ASTRONOMY | MSc THESIS: DETECTABILITY AND OCCURRENCES IN CARMENES

2019-2020 | University of Granada (Spain)

BSC IN PHYSICS | BSc THESIS: ASTRONOMICAL BOUNDS OF PRIMORDIAL BLACK HOLES

2014 – 2019 | University of Salamanca (Spain)

AWARDS & FUNDING

ESO-GARCHING VISITOR PROGRAMME

- Awarded 3-month funding to study the radial distribution of ~ 80 clusters at intermediate redshift $z \sim 0.2 - 0.8$ and design a follow-up program to characterise AGN membership of these clusters.

PHD MARIE CURIE RESEARCH FELLOW

- Early Stage Researcher of the H2020 Marie Skłodowska-Curie Innovative Training Network “Big Data analytics for Black hole Evolution Studies” ([BiD4BEST](#), grant agreement No 860744, PI Francesco Shankar)

INTERNATIONAL NETWORK & COLLABORATIONS

FORNAX

2025-current

- Builder of X-ray cluster and AGN catalogues of Fornax Euclid deep field.
- Moderator of the ESA-datalabs & Slack channel
- LOC of Annual meeting 2025

300 PROJECT

2025-current

- Explore the incidence of AGN in clusters

EUCLID

2022-current

- Active member of the SWG for Clusters of Galaxies and Galaxy evolution.

BID4BEST

2020-2024

- Early stage researcher of the H2020 Marie Skłodowska-Curie Innovative Training Network “Big Data analytics for Black hole Evolution Studies” ([BiD4BEST](#), grant agreement No 860744).

CARMENES

2019-2020

- [CARMENES](#) is an experiment designed for the discovery of exoplanets in the habitable zone of M-dwarf stars using the radial velocity method based on optical and near-infrared spectroscopy. I studied the occurrence of exoplanets in CARMENCITA (CARMENES sample of M-dwarf stars) for my MSc thesis.

STUDENT SUPERVISION

- Scientific and technical assistance of the MSc student Sylvain Heutier. Now a PhD student at the [IAP](#). (Principal supervisors: Marguerite Pierre & Francois Lanusse)
- Co-supervision of Marios Kouzis (undergraduate student) during his visit in a research program of 3 months in the National Observatory of Athens, currently ESR student of [TALES](#) EU Doctoral Training Network. (Principal supervisor Antonis Georgakakis)
- Co-supervision of the bachelor thesis of Antonis Fokas. (Principal supervisor Antonis Georgakakis)

SEMINARS & CONFERENCES

- 2025/07/01-04 “AGN contamination on cluster detection” in Madonna di Campiglio, Italy (contributed talk)
- 2025/07/01-04 “Brief introduction to ASTROMATCH” in Madonna di Campiglio, Italy (contributed talk)
- 2025/07/01-04 “Pipeline and source list of the central FornaX square degree” in Madonna di Campiglio, Italy (contributed talk)
- 2025/05/13 Postdoc seminar (**invited**, CEA Paris-Saclay, Paris)
- 2024/10/23 UAM seminar (**invited**, Universidad Autonoma de Madrid, Madrid)
- 2024/03/27 ESO cheese & wine (**invited**, ESO, Garching)
- 2024/02/06-08 “Final conference Bid4BESt” in Southampton, United Kingdom (contributed talk)
- 2023/09/25-29 “A journey through galactic environments” in Porto Ercole, Italy (Poster)
- 2023/06/19-20 “CAVITY 3rd annual meeting” in Granada, Spain (contributed talk)
- 2023/05/22 Joy of Hydrosims (**invited**, USM/LMU, Munich)
- 2023/02/06 IAC AGN talks (**invited**, IAC, Tenerife)
- 2023/02/02 Theoretical physics and astronomy group (**invited**, UGR, Granada)
- 2023/01/30-02/01 “VI Meeting of AGN Research in Spain in the Era of the New Observatories” in Granada, Spain (contributed talk)
- 2022/11/25 FAME (**invited**, INAF, Bologna)
- 2022/09/28-29 “What Drives the Growth of Black Holes” in Reykjavík, Iceland (poster + spotlight talk)
- 2022/09/05-09 “XV Reunión científica de la Sociedad Española de Astronomía” in Tenerife, Spain (poster)
- 2022/07/16-24 “44th COSPAR Scientific assembly” in Athens, Greece, Symposium E1.7 “Properties and Evolution of Active Galactic Nuclei in Galaxy Clusters” (contributed talk)
- 2022/06/27-07/01 “European Astronomical Society annual meeting 2022” in Valencia, Spain, Session S5e: “Towards the next generation of X-ray surveys with Athena” (contributed talk)
- 2022/07/06 AGN coffee (**invited**, ESO, Garching)
- 2022/06/21 ESO lunch talk (**invited**, ESO, Garching)
- 2022/05/30 Southampton Monday Seminar (**invited**, University of Southampton, Southampton)
- 2022/05/12 Kaffeerunde (**invited**, MPE, Garching)

OUTREACH

- 2022-today Podcast “[Pilgrims of the Kosmos](#)”. Topic: Talk with scientists at different stages.
- September 2022 Article in “[Astronomia](#)”. Topic: AGN (Spanish)
- July 2022 Hellenic astronomical society: [HIPPARCHOS | Volume 3, Issue 5 pages 17-21](#) Topic: AGN (co-author)
- 4 December 2020 Outreach talk about exoplanets “[Radial velocities](#)” Topic: Exoplanets (Spanish)
- Summer 2019-2021-2022 Support astronomer in [Centro Astronomico de Tiedra](#). Topic: Amateur astronomy

LIST OF PUBLICATIONS

[ADS Library](#)

- **Muñoz Rodríguez, I.**, et al. (in prep). “Impact of AGN on X-ray cluster detection.”
- **Muñoz Rodríguez, I.**, et al. (in prep). “The cluster selection criteria in FornaX”
- **Muñoz Rodríguez, I.**, et al. (2024). “Scrutinising evidence for the triggering of Active Galactic Nuclei in the outskirts of massive galaxy clusters at $z \sim 1$ ”. Monthly Notices of the Royal Astronomical Society. DOI:10.1093/mnras/stae1465
- **Muñoz Rodríguez, I.**, et al. (2023). “Cosmic evolution of the incidence of Active Galactic Nuclei in massive clusters: Simulations versus observations”. Monthly Notices of the Royal Astronomical Society, 518(1), 1041-1056. DOI:10.1093/mnras/stac3114
- Fu H., et al. (**Muñoz Rodríguez, I incl.**) “ The role of black hole feedback on galaxy star formation and the degeneracy with halo quenching”. A&A, 704, A244. doi:10.1051/0004-6361/202556480

- López I. E., et al. **(Muñoz Rodríguez, I incl.)** “A CIGALE module tailored (not only) for low-luminosity active galactic nuclei.” *Astronomy and Astrophysics*, vol 692, 2024. DOI:10.1051/0004-6361/202450510
- Fu H., et al. **(Muñoz Rodríguez, I incl.)**. “Unveiling the (in)consistencies among the galaxy stellar mass function, star formation histories, satellite abundances and intracluster light from a semi-empirical perspective.” *Monthly Notices of the Royal Astronomical Society*. DOI:10.1093/mnras/stae1492
- Musiimenta B., et al. **(Muñoz Rodríguez, I incl.)**, “Ionised AGN outflows in the Goldfish galaxy: The illuminating and interacting red quasar eFEDSJ091157.4+014327 at $z \sim 0.6$ ”, *Astronomy and Astrophysics*, vol. 687, 2024. DOI:10.1051/0004-6361/202449283.
- Amado, P. J., et al. **(Muñoz Rodríguez, I incl.)**, “The CARMENES search for exoplanets around M dwarfs. Two terrestrial planets orbiting G 264-012 and one terrestrial planet orbiting Gl 393”, *Astronomy and Astrophysics*, vol. 650, 2021. DOI:10.1051/0004-6361/202140633.
- Sabotta, S., et al. **(Muñoz Rodríguez, I incl.)**, “The CARMENES search for exoplanets around M dwarfs. Planet occurrence rates from a subsample of 71 stars”, *Astronomy and Astrophysics*, vol. 653, 2021. DOI:10.1051/0004-6361/202140968.

REFERENCES

- **Dr. Antonis Georgakakis**, Research director, National Observatory of Athens. PI of TALES.
✉ age@noa.gr
- **Dr. Marguerite Pierre**, Permanent researcher, CEA Paris-Saclay. PI of FornaX, XMM-XXL, XMM-LSS.
✉ marguerite.pierre@cea.fr
- **Prof. Francesco Shankar**, Profesor of Astrophysics, University of Southampton, PI of Bid4Best
✉ f.shankar@soton.ac.uk